

Class 30006 – Financial Markets and Institutions
Università Commerciale Luigi Bocconi
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Financial Regulation (Chapter 18)



Chapter Preview

- ✓ The financial system is one of the **most heavily regulated** industries in our economy
- ✓ Today and next class
 - we will develop an economic analysis of ...
 - **why** regulation of banking takes the form that it does
- ✓ Again two key concepts of Asy Info (*Adverse Selection* and *Moral Hazard*) provide the rationale for government regulation
- ✓ Regulation: does it always work?
 - No! actually, can have perverse **effects** ...
 - increasing both *moral hazard* and *adverse selection*

List of Topics (Chapter 18)

Ten areas of government regulation

1. Government Safety Net

- Deposit Insurance
- Lender of Last Resort

2. Capital Requirements

- Basel I, II and III

3. Restrictions on Asset Holdings

4. Assessment of Risk Management

5. Macro-prudential vs. Micro-prudential

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6. Prompt Corrective Action

7. Chartering and Examination

8. Disclosure Requirements

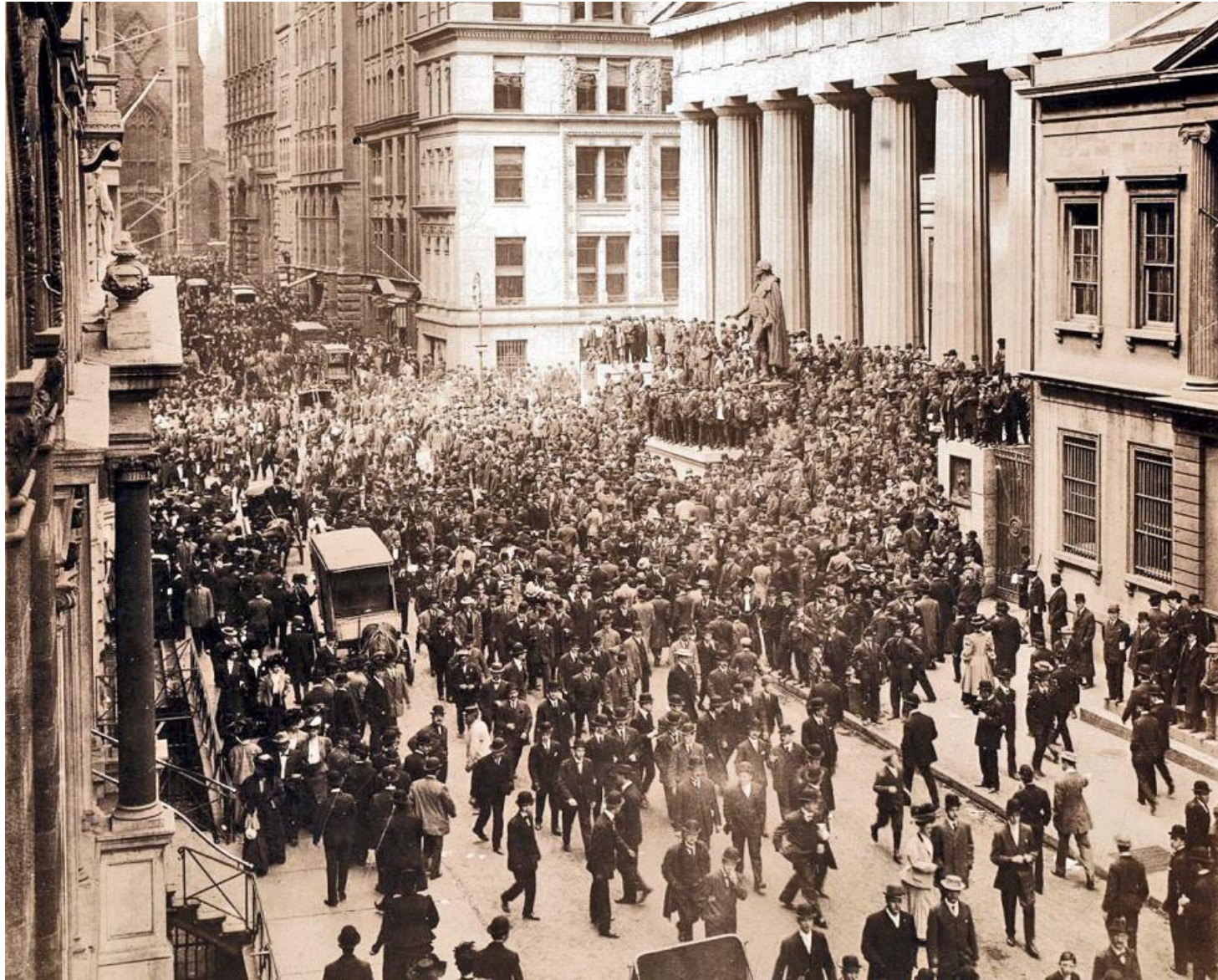
9. Consumer Protection

10. Restrictions on Competition



Study these on
your own in
the book

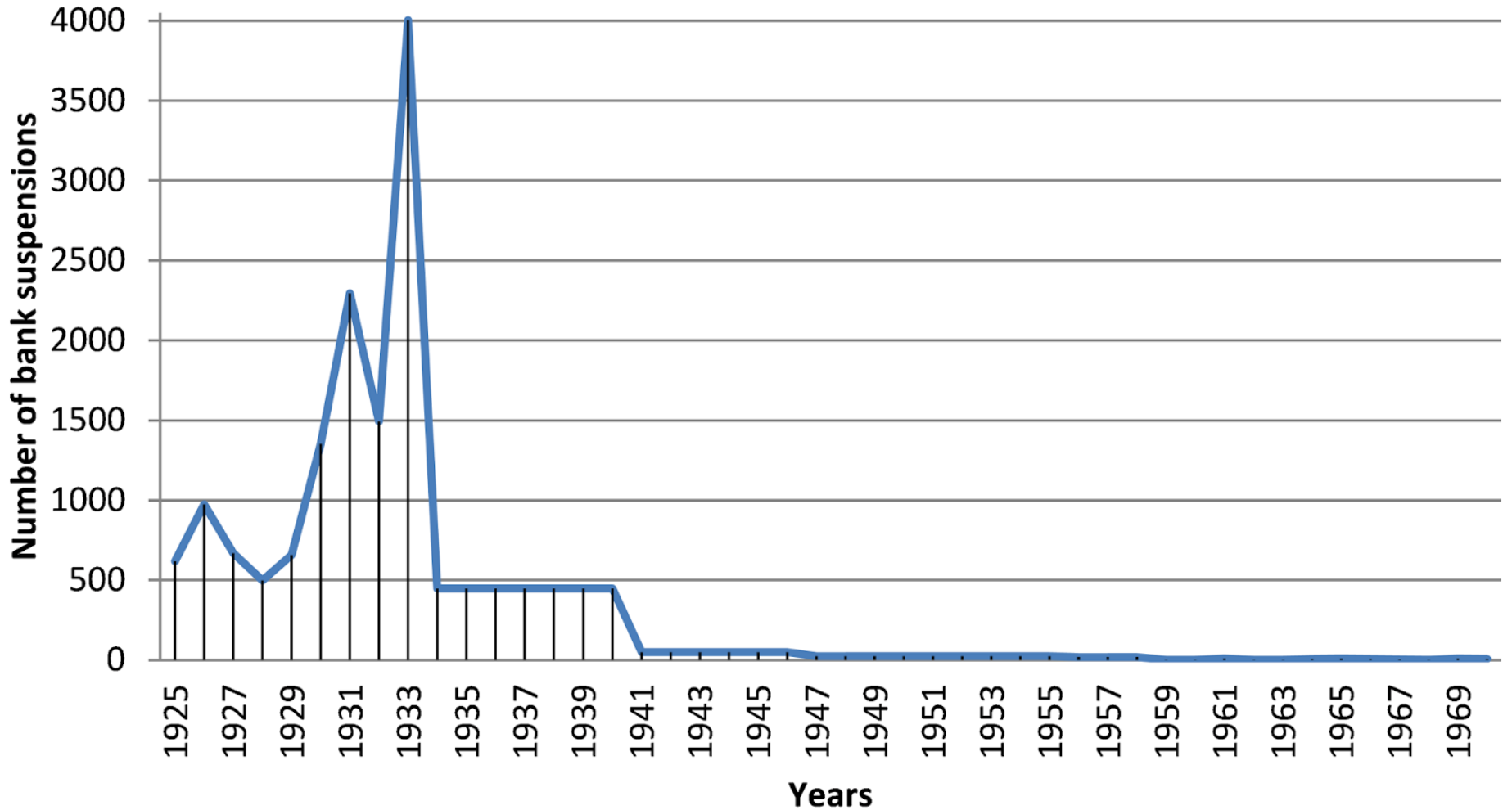
Panic of 1907: Failure of Knickerbocker Trust



Government Safety Net: Deposit Insurance

- ✓ Widespread bank failures were quite common in the 19th and early 20th century:
 - 1819, 1837, 1857, 1873, 1884, 1893, 1907, 1930-33
 - 10,000 banks failed during the Great Depression of 1930-33!
- ✓ The number was drastically reduced after 1934
 - the US government introduced **Deposit Insurance (DI)**
 - guarantee depositors' money (up to a certain limit) if the bank defaults

US Bank Suspensions 1925–1970



Bank suspension: bank temporarily suspends withdrawals to stop a bank run

Government Safety Net: Deposit Insurance

Why does **DI** prevent bank failures?

Because it prevents **bank runs**!

- ✓ A *bank run* is a **systemic event** where many depositors withdraw their money at the same time
 - we saw in Lect. 15 that banks can sustain some deposit outflows if they have *ample reserves*
 - once reserves are depleted, they need to sell other assets to raise money (or borrow from others)
 - but many assets are illiquid (loans)!
- ✓ A perfectly **solvent** bank may go bankrupt because of a **liquidity** problem created by **massive withdrawals**

Government Safety Net: Deposit Insurance

Suppose DI does not exist

Suddenly, 5% of all banks become insolvent (losses)

- ✓ Depositors are **unable to tell** **bad** and **good** banks apart (*Asy Info* between creditors & banks)
 - remember: loans are private ...
 - ... thus, banks' creditors and depositors do not know the quality of the loan portfolio! (difficult to monitor)
- ✓ Bank runs at **both** **bad** and **good** banks:
 - ✓ failure of one hastens failures at other banks (***contagion effect***)
 - ✓ “*First-come, first-served*” model gives incentives to withdraw first

Government Safety Net: Deposit Insurance

- ✓ With DI, **no incentive** to withdraw at first sign of weakness from banking system
- ✓ OK, but why should the government prevent bank failures and not failures of other firms?

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Because **banks are different**

Government Safety Net: Deposit Insurance

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- ✓ OK, but why should the government prevent bank failures and not failures of other firms?

Because **banks are different**

- ✓ A bank failure has consequences that are more **socially undesirable** than any other business
 - people lose their savings (less investment & growth) and their jobs
 - contagion effects to other banks (a rumor can be enough to trigger bank runs)
 - borrowers lose access to credit
 - information on borrower's creditworthiness is lost
- ✓ Financial intermediation in the economy may be disrupted!

Not just from the old days...

Friday 14 September 2007. Amount withdrawn: $\approx$ £1bn in 8 hours



In other countries they don't even let you run on your bank!

Withdrawal limit for Greeks summer **2015**: €60 per day, €420 per week



Bank Runs Yesterday and Today

- ✓ Bank runs have been around for as long there have been banks...
 - bank failures of [Peruzzi and Bardi](#) banks in [Medieval Florence](#) (optional)
- ✓ Last big crisis of the 2007: at the heart of it, was a **run on the shadow banking system** ([ABCP](#), repo market, MMMFs...)
- ✓ Not depositors lining up in front of the ATM ...
... but sophisticated creditors not rolling over funds
- ✓ “Modern” version of bank runs
 - read [Gorton and Metrick \(2012\)](#) on this if interested

Exercise: Deposit Insurance

Suppose you have a deposit of \$125,000 at bank X

You hear that there's a 50% chance that bank X is insolvent. DI limit is \$250,000

Q1: do you run on bank X?

Q2: what if your account is \$500,000?

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A2: \$250k not covered: **withdraw** \$250K

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✓ Of course, for the scheme to work *depositors* need to know about the insurance

- they get to know more about it when there's a crisis
- but not always: in [Italy](#) only 30% of households knew about it even in 2009

Government Safety Net: Deposit Insurance Implementation

- ✓ *Federal Deposit Insurance Corporation (FDIC)*
 - Government agency in the US that guarantees deposits up to \$25k
 - \$100k before 2008, but *unlimited* in case of [Signature Bank and SVB](#) in 2023
 - insurance fund paid by banks themselves (like an insurance premium). Similar system in other countries (EX: Italy $\leq$ €100k)
 - if money runs out (there's not enough to pay all deposits at all banks, of course), the government pays
- ✓ DI can stop bank failures with no need for paying \$:
 - it prevents runs by acting on **expectations**
- ✓ DI stops
 - not only **individual** risk of loss in case the bank fails
 - but also **aggregate** risk: bank runs from happening in the first place!

Deposit Insurance and Moral Hazard

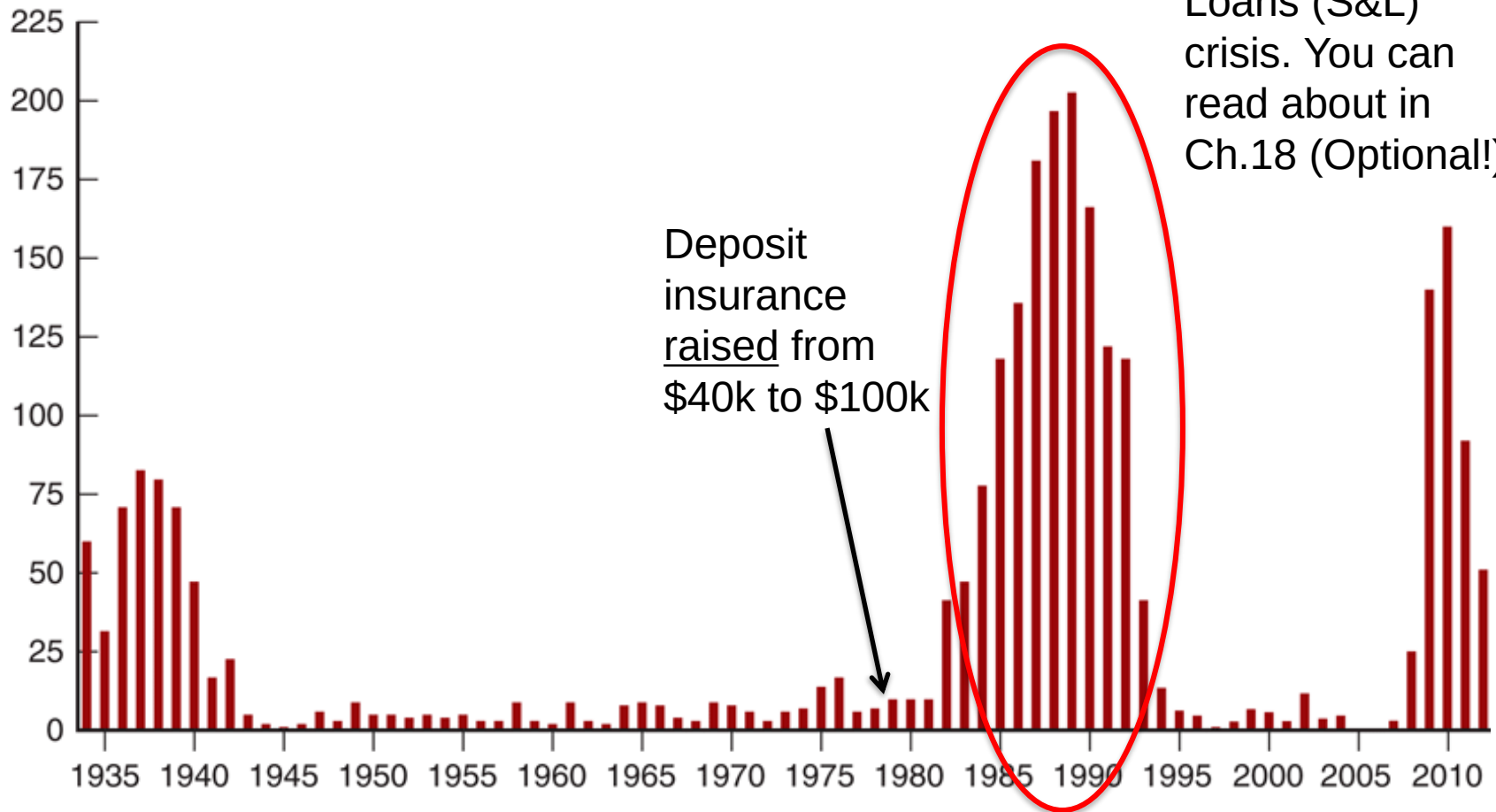
- ✓ DI is great! It can stop bank failures by stopping runs!
- ✓ Well...it does have a cost: **lack of market discipline**
- ✓ Problem:
 - if depositors' money is insured by the government
 - they will not worry about bank solvency
- ✓ Banks covered by DI face less scrutiny from creditors. They *gamble* with depositors' money and take on more risks:
moral hazard
- ✓ Logic:
 - *all gains to private owners*
 - *all losses to the public: wow!*

Deposit Insurance and Moral Hazard

- ✓ Suppose a bank invests in very risky assets
 - thus, they can pay an interest rate of 15% to its depositors compared to 0.5% that other banks pay on average
- ✓ Should you deposit \$250,000 at this bank?
 - with no deposit insurance: probably not...
 - with deposit insurance: yes, why not?
- ✓ The presence of DI can remove the disciplining effect of bank runs on bank managers and shareholders
 - especially problematic in countries with weak institutions (efficient enforcement, etc...)
- ✓ **Moral hazard** is **why** *deposit insurance* is **limited** in scope (<\$250k or <\$100k, etc...)

Bank Failures in the US, 1934-2012

Number of Bank Failures



Source: www.fdic.gov/bank/historical/bank/index.html.

Optional (Latest data [here](#), historical database [here](#))

Government Safety Net: Too-big-to-fail

- ✓ Another problem with the government safety net is the creation of the “**too big to fail**” problem:
 - some FIs become so large ... too big for the gov.t to let them fail (cost would be tremendous!)
 - markets may think that the government will always bail out (rescue) such a bank, at any cost to the taxpayer ...
 - FI know this and take advantage: they grow in size and take on excessive risks (moral hazard)
- ✓ First time term was used: *Continental Illinois* in 1984
 - **all deposits** (even $\geq$ insurance) and **all bond holders** were fully repaid: it set a dangerous precedent!
- ✓ **Too-connected-to-fail**
 - similar issue arises if the bank is *too tightly connected* to other FI's
 - contagion after a bank failure would have disastrous effects

Government Safety Net: Other Forms

- ✓ Deposit Insurance is not the only form of the government safety net

- ✓ ***Implicit guarantees*** to support banks in cases of failure are another common form

- ✓ Sometimes the central bank stands ready to provide cheap lending to banks (acting as the “**lender of last resort**”)
 - we saw this with Liquidity Management
 - Fed *discount window*, with higher rate than *Fed Funds*
 - during Financial Crisis Fed helped directly troubled FIs (“QE”)

Regulating FI: Beyond Deposit Insurance

- ✓ Government safety net
 - necessary for stability of banking system (prevents bank runs)
 - but it creates moral hazard

This motivates further *regulation*
to prevent **excessive risk taking** by FIs

2. Capital Requirements

3. Restrictions on Asset Holdings

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Capital and Returns

Question motivating capital requirements

- ✓ Suppose you have to invest all your (and your parents') wealth in two assets:
1. Asset **A** has a return of 5% for sure
 2. Asset **B** has a return of +20% with 80% probability, and a return of -80% with 20% probability

Q: Which asset would you choose?

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A: Asset B has an expected return of

$$E(\text{Ret}_B) = 20\% \times 0.8 + (-80\%) \times 0.2 = 0$$

while asset A: $\text{Ret}_A = 5\%$

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while asset A: $\text{Ret}_A = 5\%$

So, if you had to buy the asset with your own money, you would probably prefer asset A over asset B

(this is true even if risk-neutral. Only a risk lover may prefer B to A)

Capital and Returns

- ✓ Is the same true for a bank?
 - same data: $E(R_A) = 5\%$, $E(R_B) = 0$
 - similar to private investor: suppose fully capital financed and no other costs nor taxes (so here net income=net interest income)

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Asset A Bank

Assets	Liabilities
Asset A 100	Capital 100



$ROE_A =$

Asset B Bank

Assets	Liabilities
Asset B 100	Capital 100



$ROE_B =$

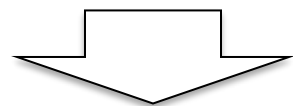
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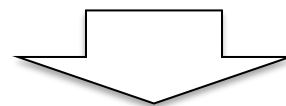
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$$ROE_A = 5\%$$

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Assets	Liabilities
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$$ROE_B = 0\%$$

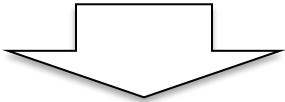
- A bank fully equity financed is like an investor using its own money

Capital and Returns

- ✓ What if the bank uses 96 of deposits and 4 of capital?
 - Deposits pay 4% in interest

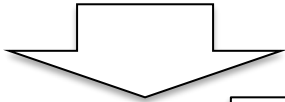
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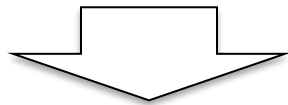

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Capital and Returns

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Asset A Bank

Assets	Liabilities
Asset A 100	Deposits 96
	Capital 4



$$ROE_A = 29\%$$

Asset B Bank

Assets	Liabilities
Asset B 100	Deposits 96
	Capital 4



$$ROE_B = 303.2\%$$

Why?

Capital and Returns

✓ Net income_A =
so ROE_A =

✓ If asset B has a high return (+20%), then
Net income_B =
so ROE_B =

Q: What is ROE if asset B has a negative return?

Q: Then what is the expected ROE_B overall?

Capital and Returns

✓ Net income_A = \$5 – (4% × \$96) = \$1.16
so ROE_A = 29%

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Capital and Returns

- ✓ Net income_A = \$5 – (4% × \$96) = \$1.16
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- ✓ If asset B has a high return (+20%), then
Net income_B = \$20 – (\$96 × 0.04) = \$16.16
so ROE_B = 404%

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A: Negative, the bank is insolvent (ROE = –100%);

it should be: Net Inc._B = [–\$80 – (\$96 × 0.04)] = –\$83.8, ROE_B = –2095%
but it is **only –100%** because of *limited liability* of shareholders (they lose only \$4!)

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Q: Then what is the expected ROE_B overall?

A: $E(\text{ROE}_B) = 0.8 \times 404\% + 0.2 \times (-100\%) = \mathbf{+303.2\%}$

Capital and Returns

Q1: What type of bank do shareholders prefer?

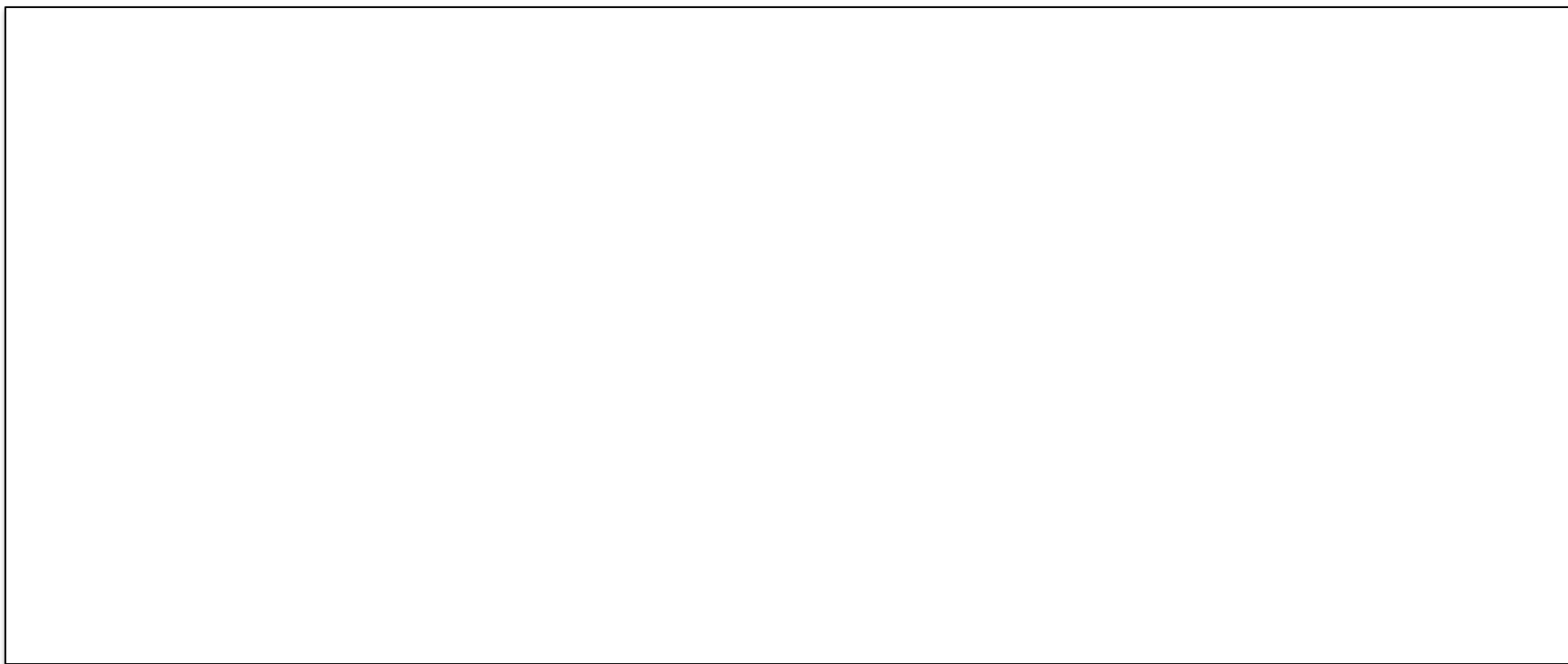
Capital and Returns

Q1: What type of bank do shareholders prefer?

A1: Obviously shareholders prefer bank B:

Bcs of **limited liability** ... they lose only capital (\$4)...

(recall: this is why $ROE = -100\% = -\$4/\4 if Asset B fails)



Capital and Returns

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✓ Now, assume deposits are fully insured. DI is free for the bank (bank pays no insurance premium)

Q2: What type of bank do depositors prefer?



Capital and Returns

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✓ Now, assume deposits are fully insured. DI is free for the bank (bank pays no insurance premium)

Q2: What type of bank do depositors prefer?

A2: Depositors should prefer bank A, but with deposit insurance they don't care: *indifferent* between bank A and bank B

Note that Deposit insurance is what allows bank B to pay only 4% in interest

Capital Requirements

- ✓ The interests of depositors and bank owners are **misaligned**
 - **less** capital the bank has, higher ROE (better for shareholders)
 - **more** capital would protect depositors (worse for shareholders)
- ✓ Solution: **Minimum Capital Requirements** (MCR)
 - idea: higher capital so if shareholders have to bear more losses, their incentive to risk-shift would diminish
 - MCR help solve the moral hazard problem induced by deposit insurance ...
 - ... FIs with a lot of capital have more to lose if they fail and are thus *less likely* to engage in *excessive risk-taking*
- ✓ Higher capital performs 2 functions:
 1. it **cushion** against bank failure
 2. it **aligns** interests of bank owners with those of depositors/creditors
 - it reduces ROE (indeed, recall $ROE = ROA \times A/E$: for a given ROA, more capital reduces ROE): lower ROE means less risky projects!

Capital Requirements

- ✓ Bank B shareholders only lost \$4, and DI covered \$76 out of the total loss of \$80 (in adverse event, $\text{loss} = 80\%$)
- ✓ In our simple example, a minimum capital requirement of **56.73%** of total assets suffices to deter risk shifting

Q: why?



Capital Requirements

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Q: why?

A: to deter risk shifting, set: $E(Ret_A) \geq E(Ret_B)$

Let Assets = NW + Deposits so, Deposits = \$100 – NW. Then:

$$\begin{aligned} & [5\% \times \$100 - 4\% \times (\$100 - NW)] \geq \\ & [20\% \times \$100 - 4\% \times (\$100 - NW)] \times 0.8 - 0.2 \times NW \end{aligned}$$

Solving you get: $NW \geq 56.73\%$

- ✓ If lower risk (project B less attractive), even a smaller amount will suffice:
 - try at home with a higher prob. of success=85%, 90%, etc...

Capital Requirements: History

- ✓ How much capital is enough? Hard to know in practice
- ✓ Through most of the 1980's, regulators used a simple minimum leverage ratio:

$$\text{leverage ratio} = \frac{E}{A}$$

- (changed notation: E=NW)
- ✓ Simple scale depending on the leverage ratio:
 - > 5%: **well**-capitalized
 - > 4%: **adequately** capitalized
 - < 4%: **under**capitalized
 - < 3%: **significantly** undercapitalized
 - < 2%: **critically under**-capitalized (to be shut down)
- ✓ If bank capital fell short of any of these thresholds, there would be restrictions to what the bank could do

Capital Requirements: History

✓ The leverage ratio had major shortcomings

- it ignored:
 - *riskiness of assets*: a risky commercial loan would require same amount of capital as safe treasury bonds
 - any *off-balance-sheet activities*: they would not incur any capital requirement at all

✓ A committee of central bankers from many nations spent a long time in Basel searching for solutions:

1. **Basel I** (1988), 26 pages
2. **Basel II** (1999), > 500 pages
implemented in 2008 for all EU (later in US)
3. **Basel III** (2013 – last update) > 500 pages & 78 equations
implementation started in 2019, ended in 2024

Capital Requirements: Basel Accord

- ✓ What is Basel all about?
 - **risk weights** applied to *all* assets and *off-balance sheet* activities
 - these weights reflect the degree of risk and are used to calculate **risk-weighted assets (RWA)**

- ✓ RWA are calculated by multiplying each asset i by its risk weight and then adding them up:

$$RWA = \sum_{i=1}^N RW_i \times A_i$$

- ✓ Bank **capital requirements** ($CapReq$) are then specified as a percentage of its RWA, rather than total assets:

$$CapReq = \frac{E}{RWA} \geq 8\%$$

- ✓ $CapReq$ are what matters to determine bank level of risk

How are the risk weights calculated?

- ✓ **Basel I** just assigned a RW to each category
 - 0% (no capital needed) on reserves and government debt of advanced economies
 - 20% on securities from *corporates* rated $\geq$ AA-
 - 50% on municipal bonds, residential mortgages
 - 100% on loans to consumers and corporations

Simple rule, but 2 problems:

- ✓ Risk weights for each asset class: too coarse and ad-hoc
- ✓ Banks engaged in **regulatory arbitrage**:
 - invest in the riskiest asset within the same risk weight class
 - Ex: would an Italian bank in 2012 buy German bond yielding 1% or Italian govt. bonds yielding +7%? They were assigned the same 0% risk-weight anyways!!

Example: Basel I

Assets			Risk-weighted
Cash	8	0%	0
Balances due from Fed	13	0%	0
Treasury bills	60	0%	0
Long-term Treasury securities	50	0%	0
Items in process of collection	10	20%	2
AA+ rated loans to other Bank	10	20%	2
Commercial loans, AAA- rated	55	20%	11
Residential mortgages	308	50%	154
Municipal bonds	110	50%	55
Commercial loans, BB+ rated	75	100%	75
Credit card loans	20	100%	20
<i>Total</i>	719	<i>Total risk-weighted assets: 319</i>	

Capital Requirements: Basel Accord

- ✓ **Basel II** changed how RWs are calculated
- ✓ Banks can use two methods:
 1. *A standardized approach* like in Basel I
 - but with **more refined buckets of risks**
 2. Or *internal model based* (**internal ratings based** or IRB) approach:
 - banks use *their own models* to determine the RWs
 - the models need to be validated by regulators
 - IRB is very complicated ... only the largest banks can afford to use the IRB

Capital Requirements: Basel Accord

- ✓ 2007–2009 crisis revealed further problems of Basel II
 - it did *not* require *enough capital* to weather the Financial Crisis (many banks were insolvent!)
 - it relied on *credit ratings* for weights — which proved unreliable during the crisis
 - *procyclical* requirement: since the RWs increase during a bad time, it requires more capital from the bank exactly when it is most short (due to write-offs)

- ✓ **Basel III** is again trying to fix these issues
 - Emphasis on quality of capital (on numerator rather than denominator)
 - *Less procyclical*: *countercyclical buffers* to be filled in good times when credit grows excessively
 - *Risk-weighted requirement* + *leverage ratio* + *liquidity ratios*
 - Liquidity ratios mean that a bank is required to hold enough high-quality liquid assets to fund cash outflows for 30 days

List of Topics (Chapter 18)

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~~1. Government Safety Net~~

- ~~○ Deposit Insurance~~
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Restrictions on Asset Holding

- ✓ There exist several other measures for reducing moral hazard, complementary to capital requirements
- ✓ Regulations **limit** the **type of assets** FI may hold
 - problem: difficult for bank stakeholders (depositors) to monitor bank
 - idea: **avoid banks hold most risky assets ...**
 - ... so moral hazard can be suppressed with *more moderate capital requirements* than if banks can just hold any risk
 - EX: US commercial banks may not hold common equity
 - Pension funds can only hold AAA securities
- ✓ Does this work?
 - in 2007–2009 Financial Crisis the risk was not in equity, rather with subprime-related securities (which were rated AAA!) ... thus, it is hard to know in advance proper rating!
 - **hard limits** may **impair efficiency** of the financial system

Assessment of Risk Management

- ✓ In earlier times, bank examiners used to only examine the quality of assets using **book values**
 - the quality of assets may be deemed acceptable at only one point in time ...

- ✓ Not longer enough now
 - banks today are able to engage in enormous risks *very fast*
 - so, balance sheet positions can **deteriorate quickly**
 - a bank that is seemingly in good shape today may be insolvent tomorrow
 - or new risks may emerge the bank is unable to deal with

Assessment of Risk Management

- ✓ Quality of assets is still important, but there is a *new trend* to shift towards
 - **monitoring risk management systems** of FI's by Supervisory Authority
 - bank examiners focus on whether the bank may take excessive risk in the near future
 - EX: check adequacy of internal policies limiting risk activity
 - EX: check quality of risk measurement and monitoring
- ✓ As part of assessing FIs' risk management, particularly important for regulators is the implementation **stress tests**
 - ✓ **stress tests** calculate the bank's losses (*i.e.* capital shortfall) under extreme circumstances
 - such as a 10% drop in the GDP or a 2-year recession
 - if a bank fails a stress test, usually a big deal!

Assessment of Risk Management

- ✓ **VaR** (*Value at Risk*) is the minimum potential loss a bank may have with a certain probability over a period of time (percentile associated to the distribution of returns) with a given level of confidence

- ✓ Examples
 - a one-day 1% VaR of \$10M:
 - a 1% probability of *at least* \$10M loss over one-day period
 - or 99% probability of losing less than \$10M in a day

- ✓ GS has an average daily 1% VaR of \$80M

Micro- Vs. Macro-Prudential Regulation

- ✓ **Micro-prudential** regulation focuses on the solvency of *individual institutions* using the tools we have seen so far
 - Stress testing, VaR, capital requirements...
- ✓ The crisis has emphasized that looking at only individual institutions is not enough:
 - the financial system is more than the sum of its parts because FI's are all *interconnected*
- ✓ **Macro-prudential** instead focuses on the solvency of the *entire financial system*
 - Financial Stability and regulation
 - “**Macro-pru**” is very difficult to do. Some indicators
 - 4ex.: look at *Credit/GDP* to spot a new crisis coming
 - 4ex.: look at *leverage cycle*, which feeds asset bubbles by offering more lending